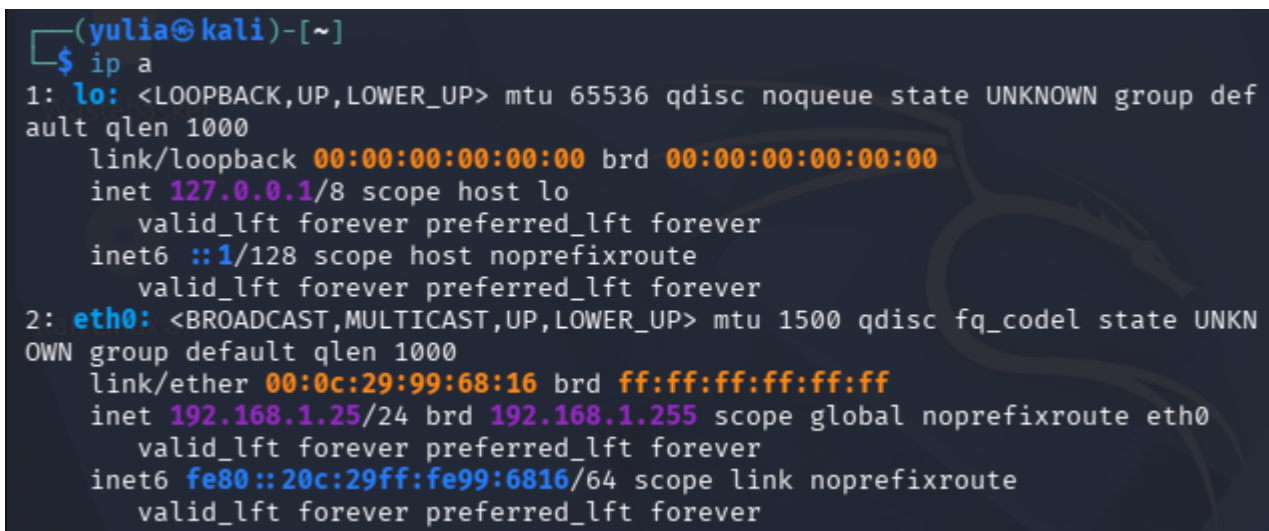


Caldera server

Operating system configurations

The file kali-linux-2025.4-installer-amd64.iso was downloaded and installed on the virtual machine in the VMware Workstation Pro. It was provisioned with 4 GB of RAM and 60 GB of disk space with the network adapter set to bridged mode. The IP address of the guest was verified with the command `ip a`.

A terminal window with a dark background and light-colored text. The prompt is '(yulia@kali)-[~]'. The command '\$ ip a' has been executed. The output shows details for two network interfaces: 'lo' (loopback) and 'eth0' (ethernet). The 'lo' interface has an IP of 127.0.0.1. The 'eth0' interface has an IP of 192.168.1.25 and a MAC address of 00:0c:29:99:68:16. The terminal text is as follows:

```
(yulia@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKN  
OWN group default qlen 1000  
    link/ether 00:0c:29:99:68:16 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.25/24 brd 192.168.1.255 scope global noprefixroute eth0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::20c:29ff:fe99:6816/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

Figure 1: IP configuration on the Kali Linux machine

Because the guest could not ping the other virtual machines, but they could ping the guest, its default gateway was configured as follows.

```
sudo nmcli connection modify "Wired connection 1" \  
ipv4.addresses 192.168.1.25/24 \  
ipv4.gateway 192.168.1.128 \  
ipv4.method manual  
nmcli connection modify "Wired connection 1" \  
    ipv4.gateway 192.168.1.128  
nmcli connection modify "Wired connection 1" \  
    ipv4.dns "8.8.8.8 1.1.1.1"  
nmcli connection down "Wired connection 1"  
nmcli connection up "Wired connection 1"
```

Caldera platform deployment

After several Caldera installations that required extensive troubleshooting, the walkthrough by Jean-Louis was followed to install Caldera 5.3.0 without any errors. The only deviation from the guide involved the ~/.zshrc configuration used by Kali Linux, rather than ~/.bashrc commands suggested by the author. The complete sequence of commands used is listed below.

```
sudo apt update && sudo apt upgrade -y

sudo apt install python3-dev git curl python3-venv npm

curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.38.0/install.sh
| bash

nano ~/.zshrc

export NVM_DIR="$HOME/.config/nvm"

[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"

[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion"

source ~/.zshrc

nvm install stable

sudo apt update

sudo apt install snapd

sudo systemctl enable --now snapd.socket

sudo snap install go --classic && sudo snap install upx

python3 -m venv venv

source venv/bin/activate

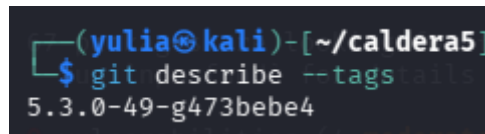
git clone https://github.com/mitre/caldera.git --recursive caldera5

cd caldera5

pip3 install -r requirements.txt

python3 server.py --insecure --build
```

To confirm the Caldera version, the output of the command `git describe --tags` was used.



```
(yulia@kali)-[~/caldera5]
$ git describe --tags
5.3.0-49-g473bebe4
```

Figure 2: Verification of the Caldera version 5.3.0 on the Kali Linux machine

Emu plugin activation

To activate the Caldera Emu plugin, the line `-emu` was added to the list of plugins in `conf/local.yml` and `conf/default.yml` files, and the script `download_payloads.sh` located in `plugins/emu` directory was executed. However, the new adversary profiles did not appear on the Caldera dashboard. To fix the issue, the following commands were issued to manually decrypt the missing files.

```
pip3 install pyminizip
```

```
python3
```

```
./plugins/emu/data/adversary-emulation-plans/sandworm/Resources/utilities/crypt_executables.py -i ./ -p malware --decrypt
```

Agent binary

In order to compile an agent for the Windows 10 target, `localhost` was changed to `192.168.1.25` in the `plugins/sandcat/gocat/sandcat.go` file. Then the instruction `G00S=windows go build -o ../payloads/sandcat.exe -ldflags="-s -w"` `sandcat.go` was executed. It did not complete as the `snap-confine` has elevated permissions and is not confined but should be error was thrown. To fix the issue, Go was reinstalled with the command `sudo apt update && sudo apt install -y golang-go`. Then the command to compile an agent was tried again; however, the same error was displayed. The Go version was verified with the instruction `which go`, and the output showed the path `/snap/bin/go`. Consequently, the command `sudo snap remove go` was issued to remove Go from snap, followed by `hash -r` to clear the terminal's lookup cache. Then the `which go` instruction was executed again and this time `/usr/bin/go` was

returned in the output. Finally, the command agent generation was attempted again, completing successfully.